

Description

This week, I learned about missing data and different imputation methods, including simple imputation using the mean, median, or mode and KNN imputation based on similar observations. We also examined a patient dataset where some zero values could represent missing data rather than genuine measurements.

Feelings

Initially, I found imputation straightforward because mean or median imputation seemed easy. However, the examples showed me that it was more complicated than I expected, so I became more cautious about choosing an appropriate method.

Evaluation

A positive part of the experience was that I considered the characteristics of the data when choosing an imputation method. I carefully examined the patient examples instead of immediately applying a method. However, deciding whether unusual values, such as zeros, were genuine or missing was difficult. Although this took more time, it helped me avoid inappropriate assumptions.

Analysis

KNN may provide more contextually appropriate estimates in some patient datasets because it uses information from similar observations. I now understand that missing-data treatment requires programming skills, statistical reasoning, and consideration of the feature's context and distribution. This is important for my development as a data scientist because poor preprocessing can lead to unreliable results.

Conclusion

Overall, my understanding of data preprocessing has improved. I learned that no single imputation method suits every dataset and that missing data should be carefully examined before further analysis. If I approached the task again, I would spend more time investigating the context of unusual values before selecting an imputation method.

Action Plan

In future, I will check each feature for unusual values, such as zeros or blank entries, and determine whether they are genuine observations or missing data. I will use `SimpleImputer` to test mean, median, and mode imputation, and apply KNN imputation when appropriate. I will then compare summary statistics and visualisations before and after imputation to check whether the chosen method has changed the data substantially.

